

D829

Digital Forensics in Cybersecurity

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A responder at a logistics company with 700 trucks arrives at a running server suspected of compromise. The server was bought from a well-known vendor three years ago. Following the order of volatility, what should be collected first?

- A. A full image of the hard drive
- B. The contents of system memory (RAM)
- C. Shut the server down, since disk evidence is always most important.
- D. The archived backup tapes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which of these sources should normally be collected last?

- A. Active network connections
- B. Contents of RAM
- C. Archived backup media
- D. The list of running processes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A 2 TB drive seized at a credit union headquarters with a marble lobby sat for six hours in an unlocked desk drawer, with no log entry, before reaching the lab. What is the main risk?

- A. A gap in the chain of custody that the other side can use to challenge the evidence.
- B. It matters only in criminal cases, never civil ones.
- C. No risk, because a later hash will match.
- D. The drive is too large to image reliably.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which record documents the chain of custody?

- A. A log of each person who handled the evidence, with dates, times, purpose, and transfers
- B. A photo of the scene at seizure
- C. The examiner's final conclusions
- D. The SHA-256 value of the image

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An examiner images a 1 TB drive from a dental office with eight treatment chairs. The SHA-256 hash of the original and the hash of the image match. What does this show?

- A. The drive contains no malware.
- B. The drive's files are encrypted.
- C. The image is a bit-for-bit copy of the original.
- D. The suspect is definitely responsible.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Weeks later, an image's SHA-256 hash no longer matches the hash recorded at acquisition. What should the examiner conclude?

- A. Hash values naturally drift over time.
- B. It does not matter if the files still open normally.
- C. The image has changed since acquisition, so its integrity cannot be relied on; re-verify against the original or a verified copy.
- D. The original drive was definitely tampered with.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An examiner at a firm that runs three labs must analyze the drive from a 2019-model laptop. What is the proper method?

- A. Boot the laptop and browse the files.
- B. Drag and drop the visible files to a USB stick.
- C. Analyze the original directly, since the firm has three labs.
- D. Attach the drive through a write blocker, make a bit-for-bit image, verify hashes, and analyze the image.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A company investigator has written consent to examine an employee's company-owned laptop. On the desk is the employee's personal phone, a brand-new model. What should the investigator do with the phone?

- A. Search it, since it was on company property.
- B. Search it, since investigators may examine any device in the office.
- C. Search it first, since new phones hold more data.
- D. Leave it unsearched, document it, and seek consent or legal authority if it is relevant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A web server in a Denver data center logs in local time (UTC-7). The firewall logs in UTC. The server shows an event at 14:30 local time. The lead investigator has 15 years of experience. What UTC time should be compared with the firewall log?

- A. The logs cannot be compared across time zones.
- B. 21:30 UTC
- C. 14:30 UTC
- D. 07:30 UTC

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A firewall log shows an event at 03:15 UTC. The matching event on a server set to UTC+2 is logged in local time. What local time should the server show?

- A. 15:15
- B. 05:15
- C. 03:15
- D. 01:15

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An examiner is writing a report for a case in which the client's attorney prefers short documents and has asked for results that support the client. How should the report be written?

- A.** State methods, tools and versions, hashes, and all relevant findings, separating facts from opinions.
- B.** Treat the report as final; it can never be supplemented.
- C.** List only conclusions and leave out methods for brevity.
- D.** Include only evidence that supports the client.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

On a Friday night, a hospital IT team of 40 finds ransomware on a file server in a third-floor rack. An administrator wants to wipe and reimage it at once to restore service by Monday. What is the best approach?

- A.** Pull the power plug and leave the server off.
- B.** Wipe and reimage now, then look for evidence.
- C.** Skip evidence, since ransomware evidence never matters.
- D.** Isolate the server from the network, capture memory and a disk image, then eradicate and restore.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A responder at a logistics company ~~with 700 trucks~~ arrives at a running server suspected of compromise. ~~The server was bought from a well-known vendor three years ago.~~ Following the order of volatility, what should be collected first?

Target word: first item to collect **Decides:** The server is running; memory is the most volatile.

A	Inverted operation	Disk data persists; imaging first risks losing memory.
B	Correct	RAM is lost when power is removed, so it comes first.
C	Absolute language	Shutting down destroys memory evidence.
D	Adjacent category	The least volatile source; collected last.

2. Answer C Concept 1 • transfer

Which of these sources should normally be collected last?

Target word: reverse: least volatile source **Decides:** Archival media barely change over time.

A	Confusion set	Changes in seconds; collected early.
B	Inverted operation	Most volatile; collected first.
C	Correct	Least volatile, so it can wait.
D	Confusion set	Disappears at shutdown; collected early.

3. Answer A Concept 2

A ~~2-TB~~ drive seized at a credit union headquarters ~~with a marble lobby~~ sat for six hours in an unlocked desk drawer, with no log entry, before reaching the lab. What is the main risk?

Target word: risk from the gap **Decides:** Six undocumented, unsecured hours.

A	Correct	Custody must be documented continuously.
B	Absolute language	Custody matters in any legal proceeding.
C	Dropped element	Drops the custody record; the hash cannot show who had access.
D	Camouflage	Capacity does not decide the custody problem.

4. Answer A Concept 2 • transfer

Which record documents the chain of custody?

Target word: reverse: from concept to artifact **Decides:** Who handled it, when, and why.

A	Correct	That is the custody record.
B	Dropped element	Documents one moment, not every transfer after it.
C	Confusion set	Findings are not a custody record.
D	Adjacent category	Shows integrity, not who handled the item.

5. Answer C Concept 3

An examiner images a 1 TB drive from a dental office with eight treatment chairs. The SHA-256 hash of the original and the hash of the image match. What does this show?

Target word: meaning of matching hashes **Decides:** Identical hashes.

A	Adjacent category	Hashes verify sameness, not cleanliness.
B	Confusion set	Hashing is not encryption.
C	Correct	Matching cryptographic hashes verify integrity of the copy.
D	Absolute language	Integrity says nothing about who did what.

6. Answer C Concept 3 · transfer

Weeks later, an image's SHA-256 hash no longer matches the hash recorded at acquisition. What should the examiner conclude?

Target word: reverse: meaning of a mismatch **Decides:** Different hash means different data.

A	Inverted operation	The same data always produces the same hash.
B	Absolute language	Changes can be invisible when opening files.
C	Correct	Any change to the data changes the hash.
D	Dropped element	Only the image was checked; the original may be fine.

7. Answer D Concept 4

An examiner at a firm that runs three labs must analyze the drive from a 2019 model laptop. What is the proper method?

Target word: correct acquisition method **Decides:** Evidence must not change.

A	Inverted operation	Bootimg writes to the drive and alters evidence.
B	Dropped element	Misses deleted data and slack space and can change metadata.
C	Camouflage	Lab count does not decide method.
D	Correct	Protects the original and preserves all data.

8. Answer D Concept 5

A company investigator has written consent to examine an employee's company-owned laptop. On the desk is the employee's personal phone, a brand new model. What should the investigator do with the phone?

Target word: scope of authority **Decides:** Consent covers only the company laptop.

A	Adjacent category	Location does not grant authority over a personal device.
B	Absolute language	Authority is limited to what consent or a warrant covers.
C	Camouflage	Storage size does not create authority.
D	Correct	The phone is outside the authorized scope.

9. Answer B Concept 6

A web server in a Denver data center logs in local time (UTC−7). The firewall logs in UTC. The server shows an event at 14:30 local time. ~~The lead investigator has 15 years of experience.~~ What UTC time should be compared with the firewall log?

Target word: normalized time **Decides:** $\text{UTC} = \text{local} - (-7) = \text{local} + 7$.

A	Absolute language	Normalizing to UTC makes them comparable.
B	Correct	$14:30 + 7 \text{ hours} = 21:30$.
C	Dropped element	Ignores the offset.
D	Inverted operation	Subtracts 7 hours instead of adding.

10. Answer B Concept 6 · transfer

A firewall log shows an event at 03:15 UTC. The matching event on a server set to UTC+2 is logged in local time. What local time should the server show?

Target word: solve for local time **Decides:** $\text{Local} = \text{UTC} + 2$.

A	Confusion set	Confuses a 12-hour reading with the 24-hour clock.
B	Correct	$03:15 + 2 \text{ hours}$.
C	Dropped element	Ignores the offset.
D	Inverted operation	Subtracts the offset.

11. Answer A Concept 7

An examiner is writing a report for a case in which ~~the client's attorney prefers short documents~~ and has asked for results that support the client. How should the report be written?

Target word: report standard **Decides:** Reports must be objective and reproducible.

A	Correct	Another examiner must be able to reproduce the work.
B	Absolute language	Reports can be supplemented when new facts emerge.
C	Dropped element	Without methods, results cannot be checked.
D	Adjacent category	That is advocacy, not objective examination.

12. Answer D Concept 8

On a Friday night, a hospital IT team ~~of 40~~ finds ransomware on a file server ~~in a third-floor rack~~. An administrator wants to wipe and reimage it at once to restore service by Monday. What is the best approach?

Target word: order of response steps **Decides:** Evidence is lost if the server is wiped first.

A	Dropped element	Loses memory and does nothing to restore service.
B	Inverted operation	Destroys the evidence before it is collected.
C	Absolute language	Evidence informs scope, recovery, reporting, and legal duties.
D	Correct	Containment and preservation come before eradication.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Order of volatility	Collect the most short-lived evidence first: CPU state and memory, then network and process state, then disk, then logs and archives.	Imaging the disk or shutting down before capturing memory.	A running system and a choice of what to collect first.
2	Chain of custody	Document every person who handled the evidence, when, where, and why; any undocumented gap invites challenges to admissibility.	Assuming a matching hash makes custody records unnecessary.	Evidence left unlogged or unsecured for any period.
3	Hash verification	Matching cryptographic hashes of original and copy show the copy is bit-for-bit identical; a later mismatch means the copy changed.	Reading a hash match as proof of guilt or of a clean system.	Two SHA-256 values compared.
4	Forensic imaging with write protection	Acquire through a write blocker, create a bit-for-bit image, verify its hash, and analyze the copy — never the original.	Bootting the suspect system or copying visible files.	“Analyze the drive” with the original in hand.
5	Legal authority and scope	Search only what a warrant, consent, or policy clearly covers; items outside scope need separate authorization.	Searching anything found nearby.	A device or data type not named in the authorization.
6	Timeline normalization	Convert every timestamp to one reference (usually UTC) before building a timeline: UTC = local time – offset.	Adding when you should subtract, or comparing raw local times.	Logs from sources in different time zones.
7	Forensic reporting	Reports state methods, tools and versions, hashes, and findings objectively, separating fact from opinion so another examiner can reproduce them.	Leaving out methods or only including evidence that helps one side.	Pressure for a shorter or more favorable report.
8	Preserve before you eradicate	In incident response, contain the system and preserve evidence (memory, images) before wiping or rebuilding.	Reimaging immediately to restore service.	Pressure to restore quickly after a compromise.

Course scope (public description): Digital Forensics in Cybersecurity examines the specialized field of digital forensics within the context of computer security. This course focuses on the techniques, tools, and processes used to investigate cybercrimes and security breaches, analyzing digital evidence to report findings in preparation for a possible legal action. It bridges the gap between theoretical knowledge and practical skills, preparing students to tackle real-world security challenges through the lens of digital forensics.